Concern Resolution Process for In-house Manufacturing

QC Story

QC STORY PROJECT

PROJECT TITLE :-

" To Eliminate concern of Glove box noise in Marazzo vehicle at CAI of SXTCF "

Mahindra and Mahindra Ltd, Nasik Plant, Scorpio –Xylo TCF.

	Plan	Actual
Start Date	**4.4.19**	**4.4.19**
Completion Date	**15.10.19**	**15.10.19**

Current Status: **Closed**

Cross Functional Team

Sr. No	Description	Name	Designation	Dept	Phone
1	**Team Leader**	Mahesh Savkar	L7-O	STCF	8888897833
2	**Mentor**	Rahul Narkar	DGM	STCF	9881080299
2	**Member 1**	Santosh Jadhav	L7-0	STCF	9689003630
3	**Member 2**	Prasad Deore	L7-0	L7-0	9764005447
4	**Member 3**	A ANURADHA	L7	MRV	9500103346
5	**Member 4**	THIRUMALASETTI RAMESH	L7-0	MRV	09840788543
6	**Member 5**	KRISHNAN BOOBALA	L7-0	MRV	919585744510

Step1. Select Subject

Brief Description : Glove box noise in Marazzo vehicle at CAI is repeat V1 concern & average 4 vehicle reported per day.

- What : Glove box noise reported in cai in Marazzo vehicle.
- Where : At CAI During rattle & wrap inspection during check & RFI Audit.
- When : Since FEB 2019.
- Who : By CAI line inspector.
- Why : Repeat issue in CAI,V1 severity concern.
- How : Avg 4 vehicles per day.

2. Justify the choice (reasons)

Mahindra

PROMISE 2019
Automotive Business

We will…

Double our revenue to ₹53,000 cr (USD 8 bn), while delivering superior financial performance

- Be the 2nd largest OEM in India by volume
- Double the overseas volume with significant presence in 10 markets

By being the most customer centric company

Brand and Customer Satisfaction
- Be among the Top 2 in Customer Satisfaction
- Be among the Top 3 in brand ranking in PV and #1 in UV & SCV
- Have 10 brands with CaPS > 50

Product and Technology
- Launch 18 products
- Be a pioneer in digital customer experience
- Develop one new disruptive business model

Crusade 2.0
- Drive brand preference through Crusade 2.0 of delivering ETR, FTR and DCE

People
- Be the #1 Great Place to Work in the manufacturing sector

Environment
- Reduce carbon and water footprint by 25%

Thus enabling our customers and stakeholders to Rise, as we progress towards being a Top 10 Global Automotive brand.

Mahindra

PROMISE 2019
Nashik Plant 1, Auto Division

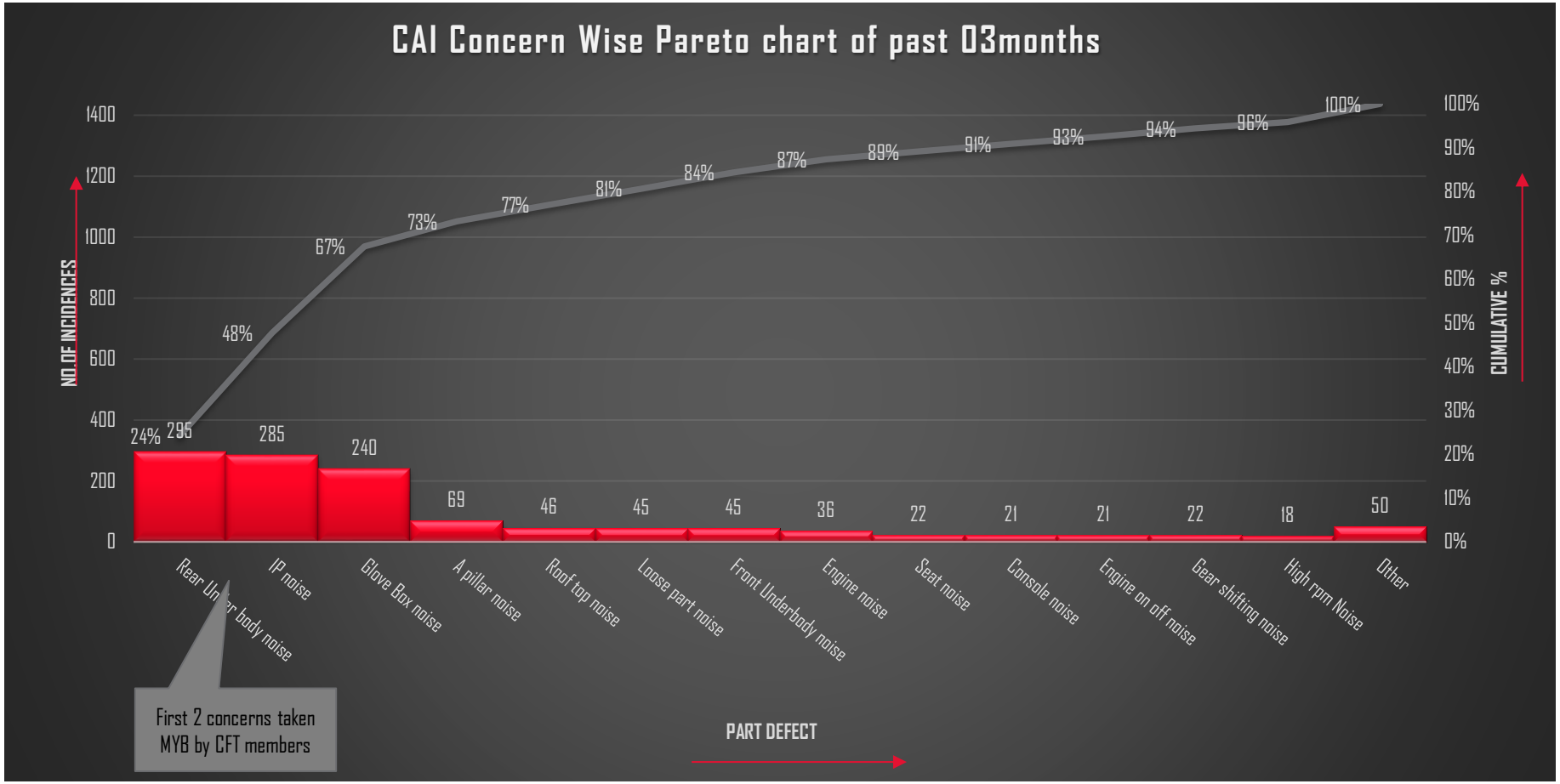
We will…

- Produce 230k vehicles in a year as per market requirement
- Fulfill 100% IO requirement for doubling the overseas volume
- Maintain cost leadership in Auto Sector by reducing revenue cost of 5% YoY
- Improve IQS for Nashik models by 25%, ranking in top 3
- Achieve CaPS > 50 for all Nashik models
- Launch 7 new products on time
- Achieve defined Crusade targets through FTR, ETR and DCE
- Achieve operational excelle[illegible] through digitization
- Achieve No.1 status for em[illegible]ee engagement in the Auto Industry
- Make officers future ready [illegible]plement new technologies coming in the next 5-10 ye[illegible]
- Reduce carbon & water foot[illegible]nt by 25%

we
nd.

Departmental Objective :-
BIQ – DPR Improvement
- FRC / FBO Improvement

2. Justify the choice



INFERENCE :-

AS PER PARETO GRAPH GLOVE BOX NOISE ISSUE IS TOP DEFECT AT CAI.

2. Justify the choice

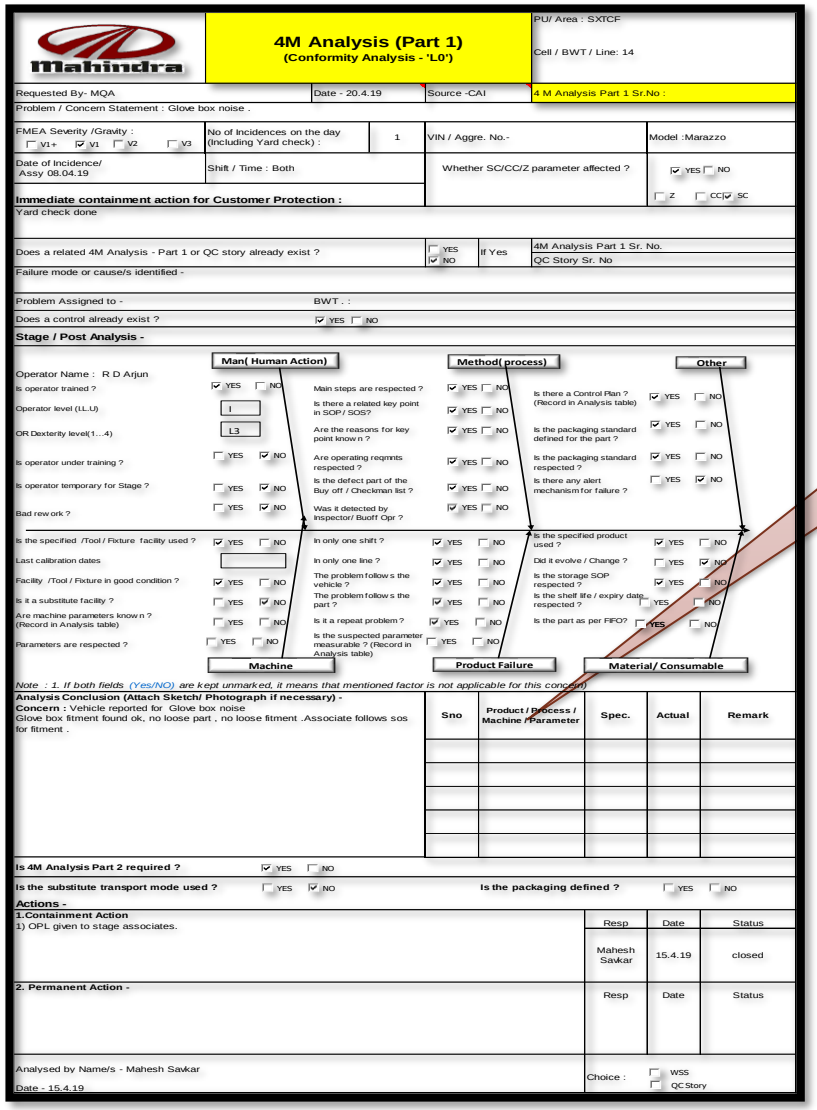
Cost of Poor Quality-Rework

Rework Cost estimation		
Description	Units	
Rework time (Approx)	Min	15
Avg Concerns/Day	Nos	4
Total Required Rework time per day	Min	60
Effective working time	Min	465
Operator required /day	nos	0.13
Approx Operator cost/ day	Rs	325
Operator cost /concern	Rs	41.94
Avg no. of working days	days	25
Operator cost /month	Rs	1048.39
Part reject/month	Nos	3.00
Part Cost	Rs	200.00
Operator Cost/year	**Rs**	**19780.65**

2.b Immediate Containment Action

- ❑ **Check point added at TRIM-2 buy off for noise checking.**
- ❑ **Check point added at FAI buy off Glove box noise checking.**

Step 3. Understanding Current Situation



4M Analysis (Part 1)
(Conformity Analysis - 'L0')

PU/ Area : SXTCF
Cell / BWT / Line: 14

Requested By- MQA | Date - 20.4.19 | Source -CAI | 4 M Analysis Part 1 Sr.No :

Problem / Concern Statement : Glove box noise .

FMEA Severity /Gravity : ☐ V1+ ☑ V1 ☐ V2 ☐ V3 | No of Incidences on the day (Including Yard check) : 1 | VIN / Aggre. No.- | Model :Marazzo

Date of Incidence/ Assy 08.04.19 | Shift / Time : Both | Whether SC/CC/Z parameter affected ? ☑ YES ☐ NO ☐ Z ☐ CC ☑ SC

Immediate containment action for Customer Protection :
Yard check done

Does a related 4M Analysis - Part 1 or QC story already exist ? ☐ YES ☑ NO | If Yes | 4M Analysis Part 1 Sr. No. / QC Story Sr. No

Failure mode or cause/s identified -

Problem Assigned to - BWT . :

Does a control already exist ? ☑ YES ☐ NO

Stage / Post Analysis -

Man(Human Action)

Operator Name : R D Arjun
Is operator trained ? ☑ YES ☐ NO
Operator level (LL/I) I
OR Dexterity level(1…4) L3
Is operator under training ? ☐ YES ☑ NO
Is operator temporary for Stage ? ☐ YES ☑ NO
Bad rew ork ? ☐ YES ☑ NO

Method(process)

Main steps are respected ? ☑ YES ☐ NO
Is there a related key point in SOP/ SOS? ☑ YES ☐ NO
Are the reasons for key point known ? ☑ YES ☐ NO
Are operating reqmnts respected ? ☑ YES ☐ NO
Is the defect part of the Buy off / Checkman list ? ☑ YES ☐ NO
Was it detected by Inspector/ Buoff Opr ? ☑ YES ☐ NO

Other

Is there a Control Plan ? (Record in Analysis table) ☑ YES ☐ NO
Is the packaging standard defined for the part ? ☑ YES ☐ NO
Is the packaging standard respected ? ☑ YES ☐ NO
Is there any alert mechanism for failure ? ☐ YES ☑ NO

Machine

Is the specified /Tool / Fixture facility used ? ☑ YES ☐ NO
Last calibration dates
Facility /Tool / Fixture in good condition ? ☑ YES ☐ NO
Is it a substitute facility ? ☐ YES ☑ NO
Are machine parameters known ? (Record in Analysis table) ☐ YES ☐ NO
Parameters are respected ? ☐ YES ☐ NO

Product Failure

In only one shift ? ☑ YES ☐ NO
In only one line ? ☑ YES ☐ NO
The problem follow s the vehicle ? ☑ YES ☐ NO
The problem follow s the part ? ☑ YES ☐ NO
Is it a repeat problem ? ☑ YES ☐ NO
Is the suspected parameter measurable ? (Record in Analysis table) ☐ YES ☐ NO

Material/ Consumable

Is the specified product used ? ☑ YES ☐ NO
Did it evolve / Change ? ☐ YES ☑ NO
Is the storage SOP respected ? ☑ YES ☐ NO
Is the shelf life / expiry date respected ? ☐ YES ☐ NO
Is the part as per FIFO? ☐ YES ☐ NO

Note : 1. If both fields (Yes/NO) are kept unmarked, it means that mentioned factor is not applicable for this concern

Analysis Conclusion (Attach Sketch/ Photograph if necessary) -
Concern : Vehicle reported for Glove box noise
Glove box fitment found ok, no loose part , no loose fitment .Associate follows sos for fitment .

Sno	Product / Process / Machine /Parameter	Spec.	Actual	Remark

Is 4M Analysis Part 2 required ? ☑ YES ☐ NO

Is the substitute transport mode used ? ☐ YES ☑ NO **Is the packaging defined ?** ☐ YES ☐ NO

Actions -

1.Containment Action
1) OPL given to stage associates.

Resp	Date	Status
Mahesh Sawkar	15.4.19	closed

2. Permanent Action -

Resp	Date	Status

Analysed by Name/s - Mahesh Sawkar
Date - 15.4.19

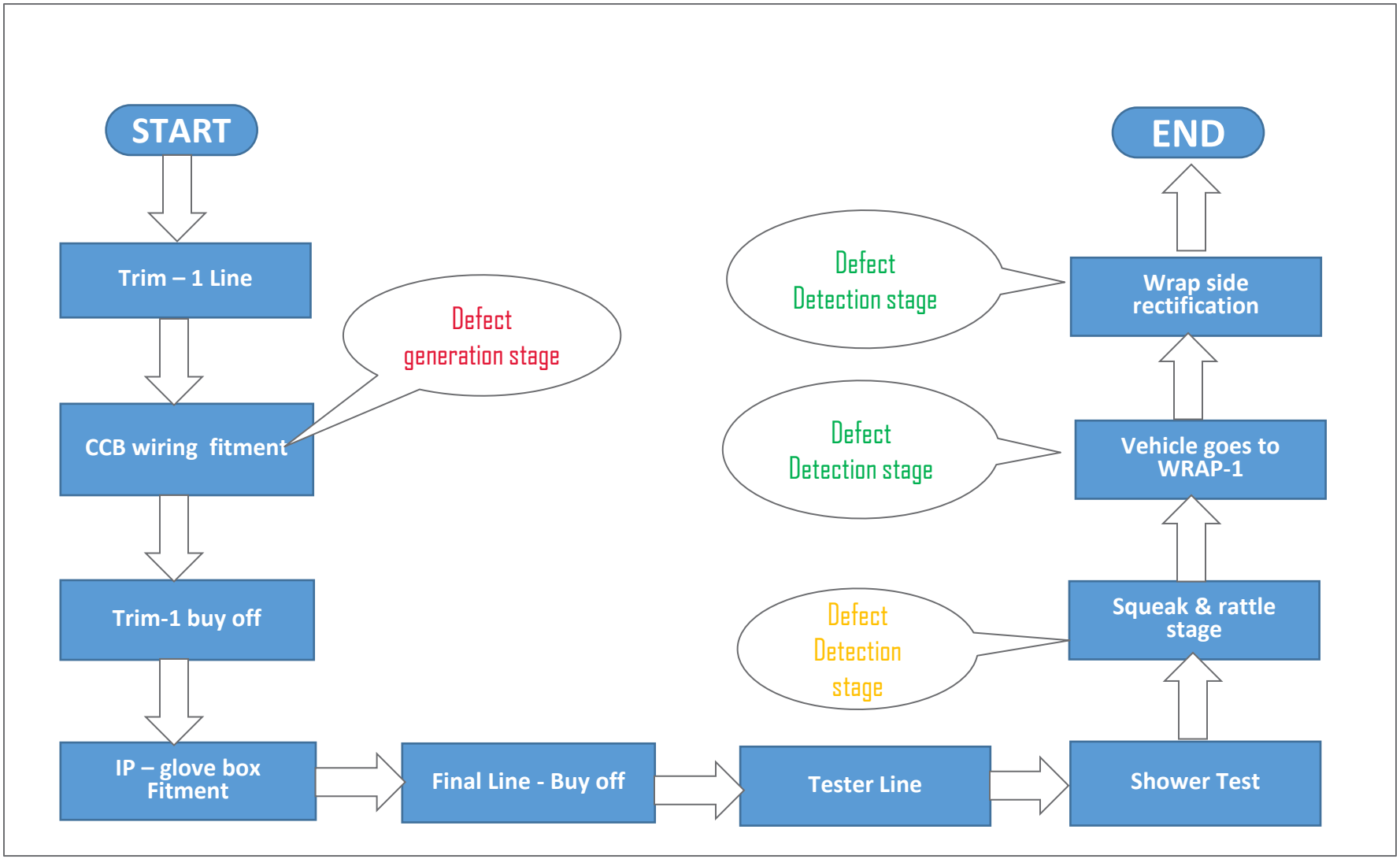
Choice : ☐ WSS ☐ QC Story

Inference from L0,
Problem follows with vehicle part,
need detail analysis.

4M analysis Good and Not so good observations	
MAN	Operator Dext. Level L3
Method	Operator aware of key point and check present at buyoff
Others	Control plan adhered on stage
Machine	Tools used are calibrated
Product failure	Problem follows vehicle
Material	Handling and shelf life of part adhered

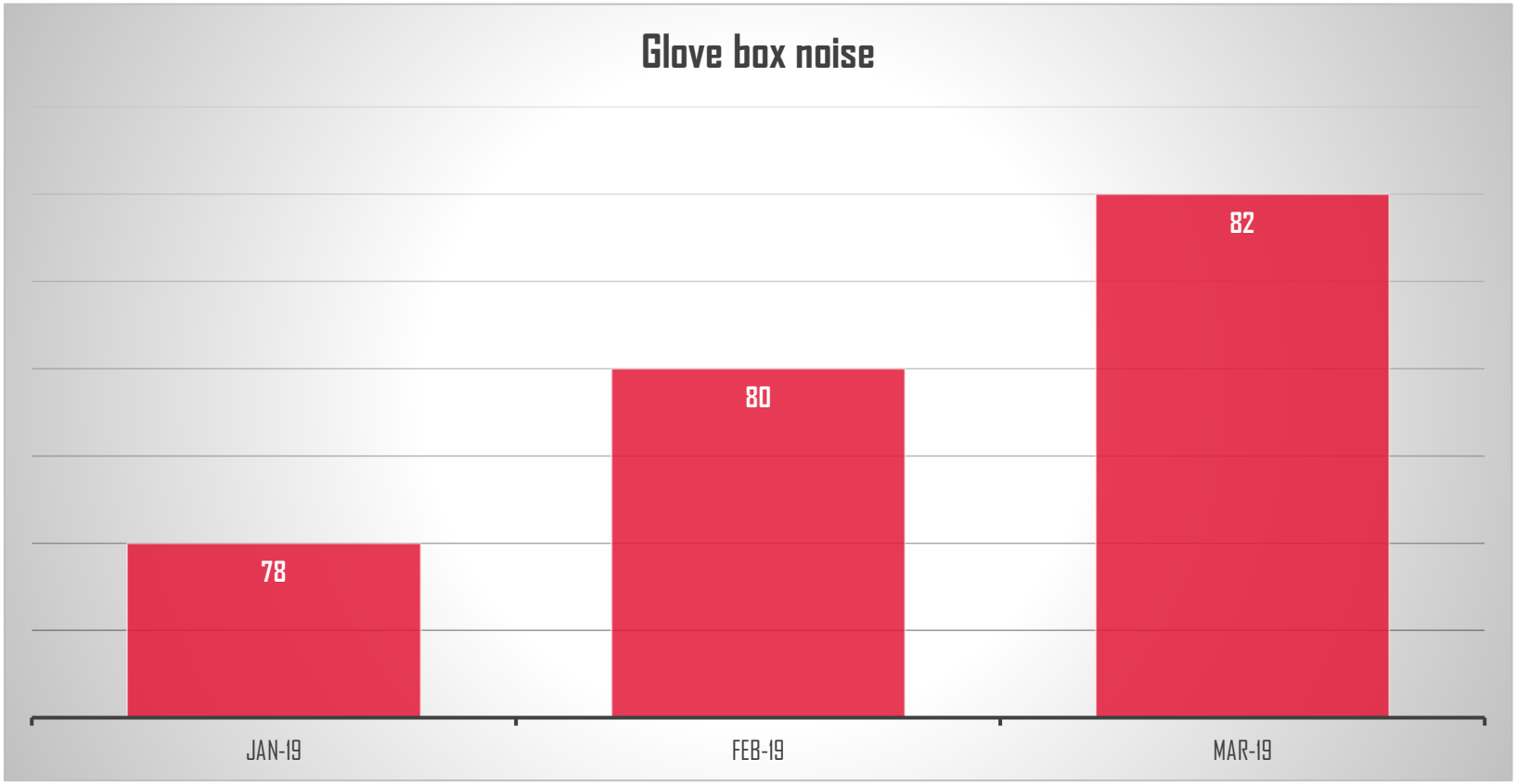
NO QUICK WINS

Step 3. Understanding Current Situation



3. Understand the current situation

Month wise Trend



INFERENCE :- As per graph Glove box noise issue consistent month on month

Step 3. Understanding Current Situation

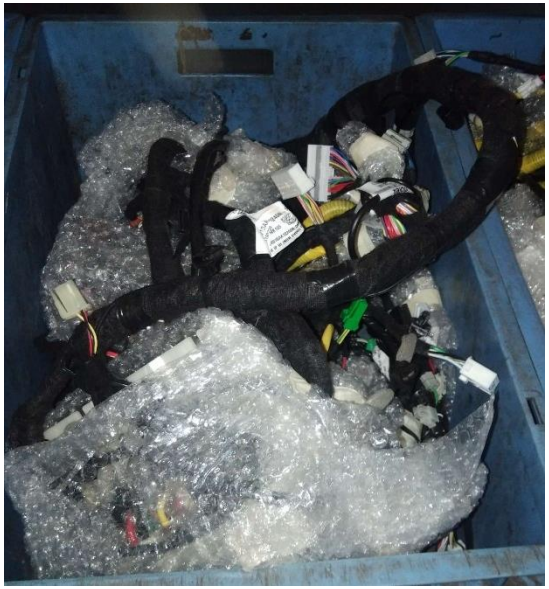
Good & Not so good observations

- Issue observed in U321 all variant.
- Issue reported in both the shifts.
- Repair time is high around 15 minutes per vehicle.
- Issue reported at CAI tester line .
- There was level 2 operators for IP fitment .
- Not all the vehicles having this issue.
- Defect follows with Vehicle.

Step 3. Understanding Current Situation

Parts Involved in the analysis:

1. CCB Wiring

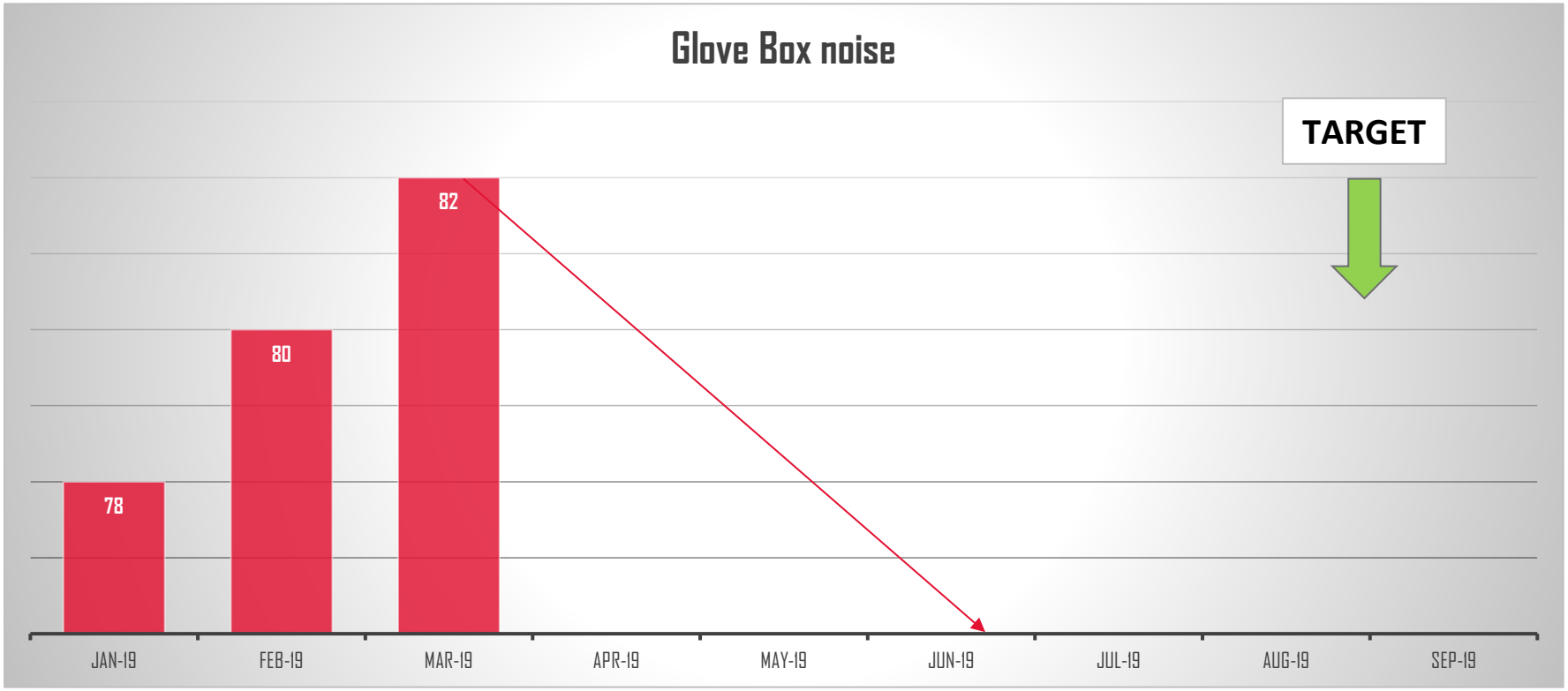


2. IP

3 . Glove box

4. Select targets

Present Level:	240 cases reported at CAI based on data of last 03 months
Target :	**Concern elimination by end of July 2019**



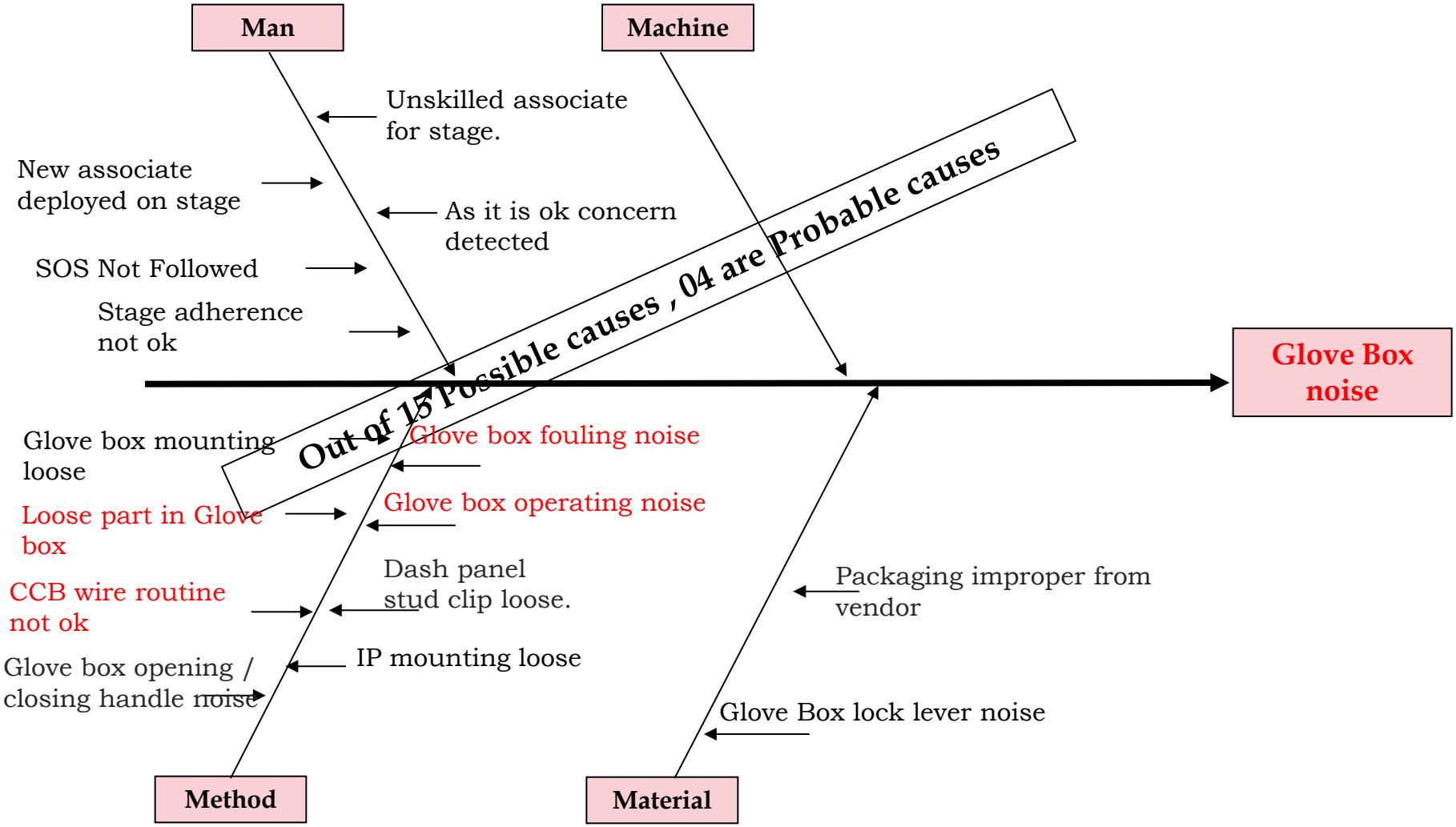
5. Analysis

List of Possible causes after brainstorming among team members

Sr. no	Possible Causes
1	New associate deployed on stage
2	SOS not followed
3	Unskilled associate for stage.
4	As it is ok concern detected.
5	Glove box mounting loose
6	Loose part in Glove box
7	Stage adherence not ok
8	CCB wire routine not ok
9	IP mounting loose
10	Dash panel stud clip loose.
11	Glove box operating noise
12	Glove box fouling noise
13	Glove Box lock lever noise
14	Glove box opening / closing handle noise

Step 5. Analysis

5.1 Identify possible causes: Fish Bone diagram



Step 5. Analysis

List of Possible causes identification in Brain storming session

Sr. no	Possible Causes	Mahesh	Santosh	Satish	Prasad	Mandar	Nilesh	Satish	Total
1	New associate deployed on stage	1	1	1	3	3	3	1	13
2	SOS not followed	3	3	1	1	1	3	1	13
3	Unskilled associate for stage.	1	3	3	1	1	1	1	11
4	As it is ok concern detected.	1	1	1	1	1	1	3	9
3	Glove box mounting loose	3	3	1	3	1	1	1	13
6	Loose part in Glove box	9	9	3	1	3	9	3	37
7	Stage adherence not ok	9	3	3	3	3	1	1	23
8	CCB wire routine not ok	3	9	1	9	9	9	3	43
9	IP mounting loose	3	1	3	3	1	1	1	13
10	Dash panel stud clip loose.	3	9	9	3	1	3	3	31
11	Glove box operating noise	3	9	9	3	3	3	9	39
12	Glove box fouling noise	3	3	9	9	9	1	3	37
13	Glove Box lock lever noise	1	9	1	9	3	3	3	29
14	Glove box opening / closing handle noise	9	3	3	1	3	3	1	23
15	Packaging improper from vendor	9	3	1	3	3	3	1	23

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 9- Significant impact.
Threshold Value for selecting cause for hypothesis is 36.

Step 5. Analysis

List of causes not considered as probable causes and reasons for not considering

SR NO	Not considered causes	Reasons
1	New associate deployed on stage	All trained associates deployed on stage
2	SOS not followed	SOS followed
3	Unskilled associate for stage.	Associate deployed as per ILU plan
4	As it is ok concern detected.	not a concern
5	Glove box mounting loose	Check point available on stage
7	Stage adherence not ok	3 x 3 matrix available on stage so issue for adherence
9	IP mounting loose	Check point available .
10	Dash panel stud clip loose.	Check Found ok
13	Glove box lock lever noise	Check Found ok
14	Glove box opening / closing handle noise	Check Found ok
15	Packaging improper from vendor	Check Found ok as per PDS

Step 5. Analysis

List of causes considered as possible causes taken for test of hypothesis.

Sr. no	Possible Causes
6	Loose part in Glove box
8	CCB wire routine not ok
11	Glove box operating noise
12	Glove box fouling noise

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	Loose part in Glove box	Visual inspection	Defective vehicle analyzed and	**Hypothesis Invalid**

Loose part in glove box is a special cause concern , checkpoint available at shower before rattle to ensure no loose part in glove box.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
2	CCB wire routine not ok	10 vehicles checked in which wire close	Observations & measurement on Not OK vehicles. (concern raise by WRAPSIDE Repairmen)	**Hypothesis valid**

Observations : -

- Wire to glove box gap less while operating .
- 5 defective vehicles analyzed – 3 vehicles got ok by wire routine change.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
3	Glove box operating noise	Visually by operating	10 vehicles checked in which all vehicles found ok for operating and locking	**Hypothesis Invalid**

Checked for glove box operating noise		
SR NO	VIN	RESULT
1	K2E49319	ok
2	K2E49064	ok
3	K2E49343	ok
4	K2E49287	ok
5	K2E49400	ok
6	K2E49167	ok
7	K2E49274	ok
8	K2E49294	ok
9	K2E49368	ok
10	K2E49283	ok

Observations : -

- Vehicles checked for operating & closing , no concern observed. Hence not a cause.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
4	Glove box fouling noise	Visually by operating checked.	10 vehicles checked in which , 3 vehicles found defective.	**Hypothesis valid**

Observations : -

- 10 vehicles checked for wire fouling.
- 3 vehicles found for wire fouling to edge resulting for glove box noise .

Checked for glove box operating noise		
SR NO	VIN	RESULT
1	K2E49458	ok
2	K2E49558	ok
3	K2E49139	not ok
4	K2E49606	ok
5	K2E49364	ok
6	K2E49717	not ok
7	K2E49436	ok
8	K2E49624	ok
9	K2E49449	not ok
10	K2E49663	ok

TEST OF HYPOTHESIS

5.3 Identify the root causes:

Sr No.	Probable cause	WHY?	WHY?	WHY?	Action
1.	CCB wire routine not ok .	LH A pillar wire sag	Routine improper		Wire routine fixed from inside LH A pillar CCB bar
2.	Glove box fouling noise	While operating glove box noise	Sag formation of wire near glove box edge	Wire fouling to Glove box edge	Fencing clip to be provided on wire.

6. Implement the corrective measures

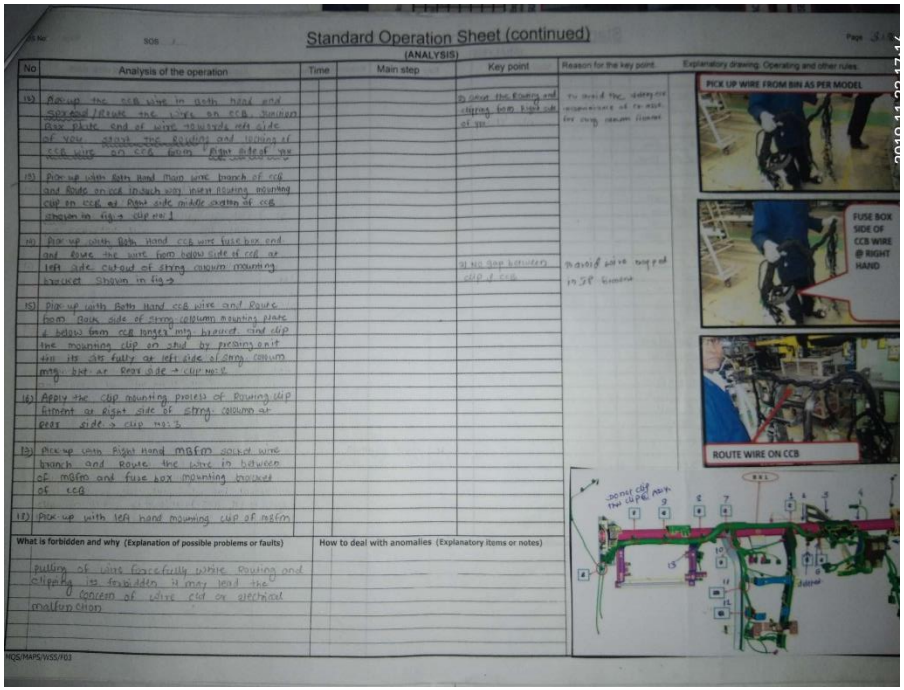
Possible Remedies Identified

Sr. No.	Root Cause	Remedy Developed	Time to implement	Cost of Improvement	Easily reversible	Compl exity	Decision
1	CCB wire routine not ok	Wire routine fixed from inside LH A pillar CCB bar.	Low	Low	Yes	Low	**Accepted**
2	Glove box fouling noise	Fencing clip to be provided on wire.	High	High	No	Low	**Accepted**

6. Implement the corrective measures

6.1 Actions Decided

Sr. No.	Root Cause	Action plan	Target Date	Responsibility	Status
1	CCB wire routine not ok	Wire routine fixed from inside LH A pillar CCB bar.	19.6.19	Prasad Deore	Closed



Standard Operation Sheet (continued)

(ANALYSIS)

No	Analysis of the operation	Time	Main step	Key point	Reason for the key point	Explanatory drawing, Operating and other rules
15)	Pick up the CCB wire in both hand and spread/route the wire on CCB. Junction Box plate end of wire towards left side of you. start the routing and locking of CCB wire on CCB bar. Right side of box			To avoid the routing and clipping from right side of you	To avoid the interference of re-route for strg. column flames	PICK UP WIRE FROM BIN AS PER MODEL
15)	Pick up with both hand main wire branch of CCB and route in such way insert routing mounting clip on CCB at front side middle portion of CCB shown in fig → clip no: 1					
16)	Pick up with both hand CCB wire fuse box end and route the wire from below side of CCB at left side cut-out of strg. column mounting bracket shown in fig →			To no gap between clip & CCB	To avoid wire rub and in SP flames	FUSE BOX SIDE OF CCB WIRE @ RIGHT HAND
15)	Pick up with both hand CCB wire and route from main side of strg. column mounting plate & bracket from CCB hanger mtg. bracket and clip the mounting clip on stud by pressing until its sits fully at left side of strg. column mtg. bkt. at rear side → clip no: 2					
16)	Apply the clip mounting process of routing clip fitment at right side of strg. column at rear side → clip no: 3					
19)	Pick up with right hand MGFM socket wire branch and route the wire in between of MGFM and fuse box mounting bracket of CCB					ROUTE WIRE ON CCB
17)	Pick up with left hand mounting clip of MGFM					

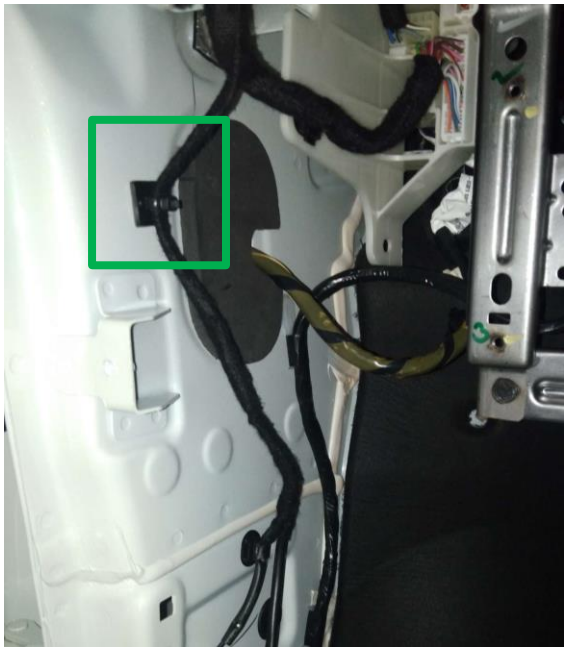
What is forbidden and why (Explanation of possible problems or faults)	How to deal with anomalies (Explanatory items or notes)
pulling of wire forcefully while routing and clipping is forbidden it may lead the concern of wire cut or electrical malfunction	

Wire routine changed to inner side , 10 vehicles checked result found ok .

6. Implement the corrective measures

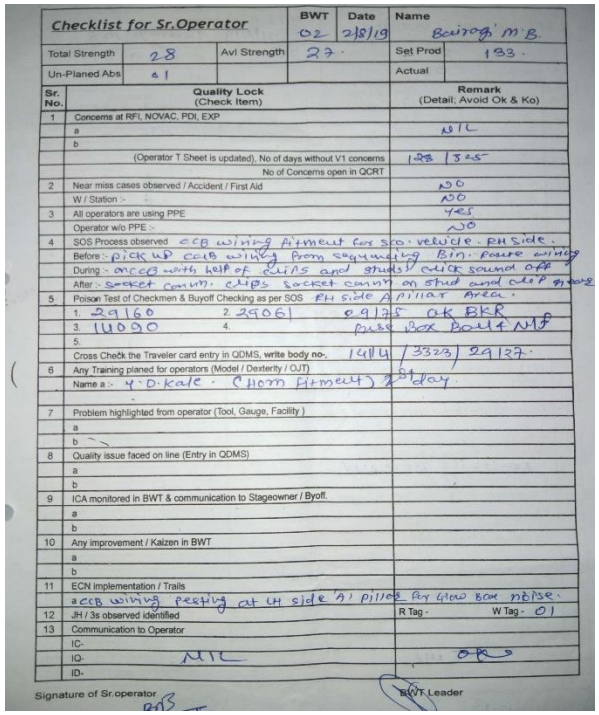
6.1 Actions Decided

Sr. No.	Root Cause	Action plan	Target Date	Responsibility	Status
2	Glove box fouling noise	Fencing clip to be provided on wire.	5.7.19	Santosh Jadhav	Closed



After clip provision 25 vehicles checked , result found ok .

S.O. CHECKSHEET UPDATED



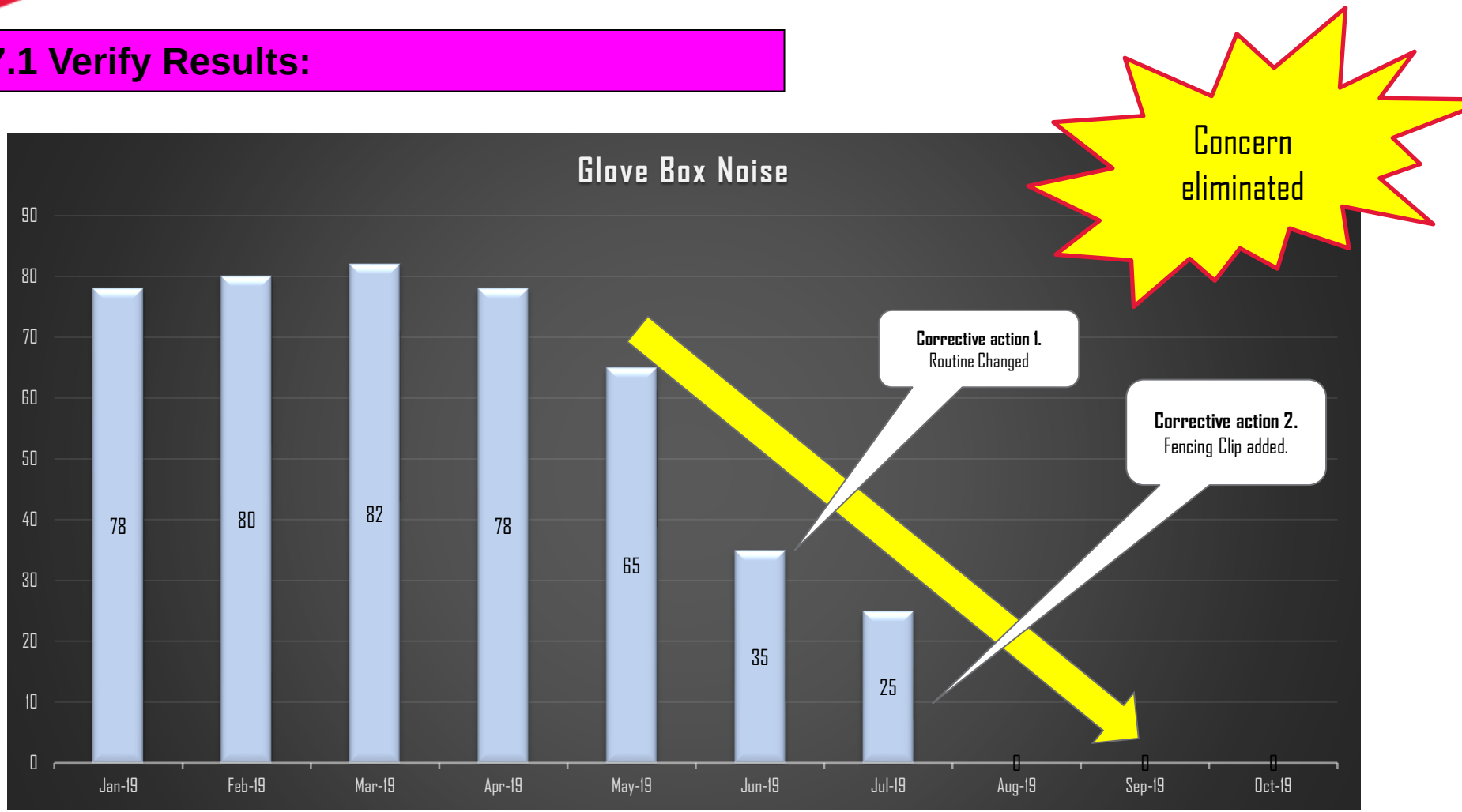
Checklist for Sr.Operator	BWT 02	Date 2/8/19	Name Borogi M.B.
Total Strength 28	Avl Strength 27	Set Prod 133	
Un-Planed Abs 01		Actual	

Sr. No.	Quality Lock (Check Item)	Remark (Detail; Avoid Ok & Ko)
1	Concerns at RFI, NOVAC, PDI, EXP	
	a	NIL
	b	
	(Operator T.Sheet is updated), No of days without V1 concerns	1-28 / 325
	No of Concerns open in QCRT	
2	Near miss cases observed / Accident / First Aid	NO
	W / Station :-	NO
3	All operators are using PPE	YES
	Operator w/o PPE :-	NO
4	SOS Process observed CCB wiring Atment for SOS vehicle. RH side.	
	Before :- pick up CCB wiring from segmenting Bin. Route wiring	
	During :- on CCB with help of clips and studs click sound off	
	After :- socket comm. clips socket comm on stud and clip [illegible]	
5	Poison Test of Checkmen & Buyoff Checking as per SOS RH side A pillar Area.	
	1. 29160 2. 29061 29175 OK BKR	
	3. 140090 4. Fuse Box Bolt NIL	
	5.	
	Cross Check the Traveler card entry in QDMS, write body no-. 1414 / 3323 / 29127.	
6	Any Training planed for operators (Model / Dexterity / OJT)	
	Name a :- Y.D. Kale. (Hom Atment) 2 day.	
7	Problem highlighted from operator (Tool, Gauge, Facility)	
	a	
	b	
8	Quality issue faced on line (Entry in QDMS)	
	a	
	b	
9	ICA monitored in BWT & communication to Stageowner / Byoff.	
	a	
	b	
10	Any improvement / Kaizen in BWT	
	a	
	b	
11	ECN implementation / Trails	
	a CCB wiring resting at LH side 'A' pillar for Glow Box noise.	
12	JH / 3s observed identified	R Tag - W Tag - 01
13	Communication to Operator	
	IC-	
	IQ- NIL	OK
	ID-	

Signature of Sr.operator — BWT Leader

7. Check & confirm the results & effects

7.1 Verify Results:



INFERENCE :- As per graph issue reduced month on month after corrective actions.

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

S N	Action Taken	Actual date	Responsibility	Status
1	SOS Updated for Key point added	19.6.19	Prasad Deore	Closed.
2	SOS updated for new fencing clip added	5.7.19	Mahesh /Prasad	Closed.
5	QA matrix Updated	5.7.19	Mahesh /Prasad	Closed
6	Awareness given through DRF	5.7.19	Prasad Deore	Closed
7	Check point added in buy off check sheet	5.7.19	Santosh Jadhav	Closed

Check & confirm the results & effects (Cont.....)

7.3 Tangible & Intangible benefits:

Tangible Benefits -

1) STR Increased by 0.5%
   i.e from 71% to 71.5%
1) Rework cost is reduced by
   approx.Rs.19780.65 /Annum.

In tangible Benefits :

1) Learning of CFT working approach
2) Gemba Culture is learned
3) Operator Morale Increased
4) Improved Customer Satisfaction.

From: DHANOKAR RAJESH
Sent: Monday, November 25, 2019 3:40 PM
To: SAVKAR MAHESH <SAVKAR.MAHESH@mahindra.com>
Subject: RE: MYB COPQ APPROVAL
Okay Approved
From: SAVKAR MAHESH
Sent: Monday, November 25, 2019 12:57 PM
To: DHANOKAR RAJESH <DHANOKAR.RAJESH@mahindra.com>
Subject: MYB COPQ APPROVAL
Sir,
Please approve the rework cost required for Glove box noise in Marazzo.

Rework Cost estimation		
Description	Units	
Rework time (Approx)	Min	15
Avg Concerns/Day	Nos	4
Total Required Rework time per day	Min	60
Effective working time	Min	465
Operator required /day	nos	0.13
Approx Operator cost/ day	Rs	325
Operator cost /concern	Rs	41.94
Avg no. of working days	days	25
Operator cost /month	Rs	1048.39
Part reject/month	Nos	3.00
Part Cost	Rs	200.00
Operator Cost/year	**Rs**	**19780.65**

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Actual date	Responsibility	Status
1	Engineering change note	12.6.19	**Santosh Jadhav**	Closed
2	Drawing	12.6.19	**Santosh Jadhav**	Closed
3	PFMEA updation	5.7.19	**Prasad Deore**	Closed
4	SOS updation	5.7.19	**Prasad Deore**	Closed
5	QA matrix Updation	5.7.19	**Vikrant Ghule**	Closed

8.2 Training:

SN	Training Identified	Actual date	Responsibility	Status
1.	Awareness given through OPL.	5.7.19	Prasad/Mahesh	Closed

8.3 Additional Audit & check

1.	Check point added in buy off check sheet	5.7.19	Prasad Deore	Closed

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

Engineering change note : U321-EN-A003-000114

EN

Mahindra Rise.

Engineering Notice Release: U321-EN-A003-000114
AD Product Development
Mahindra & Mahindra LTD(A.S)

Created Date	01-Apr-2019
Initiated Date	07-Jun-2019

Engineering Notice No : U321-EN-A003-000114

Released Date : 12-Jun-2019 15:38

Reference Change Req No : DCR042104

Location : AD-NASIK

Implementation Location : M&M-Nasik

Engineering Notice Description : EN RELEASE FOR AIS 145 CHANGES

Reason : EN RELEASE FOR AIS 145 CHANGES

Quantity : -(Regular)

Reason Code : Regulatory requirement

Priority : Urgent

Note : EN RELEASE FOR AIS 145 CHANGES

Affected Models : AU32BPES8T151A00SK ,AU32EPES7T151A00SK ,AU32FPES7T151A00SK ,AU32BPES8T151A00XH ,AU32CPES8T151A00XH ,AU32EPES8T151A00XH ,AU32EPES7T151A00XH ,AU32FPES7T151A00XH ,AU32BPES8T151A00CB ,AU32CPES8T151A00CB ,AU32EPES8T151A00CB ,AU32EPES7T151A00CB ,AU32FPES7T151A00CB ,AU32BPES8T151A00WP ,AU32CPES8T151A00WP ,AU32EPES8T151A00WP ,AU32EPES7T151A00WP ,AU32FPES7T151A00WP ,AU32BPES8T151A00GR ,AU32CPES8T151A00GR ,AU32EPES8T151A00GR ,AU32EPES7T151A00GR ,AU32FPES7T151A00GR ,AU32BPES8T151A00PD ,AU32CPES8T151A00PD ,AU32EPES8T151A00PD ,AU32EPES7T151A00PD ,AU32FPES7T151A00PD ,AU32BPES8T151A00ME ,AU32CPES8T151A00ME ,AU32EPES8T151A00ME ,AU32EPES7T151A00ME ,AU32FPES7T151A00ME ,

Is EN Re Released? :

Parent EN NO :

Life Cycle State : Released

Owner Name : K KALIRAJ (23189900)

Engineer : K KALIRAJ (23189900)

Approved By : 23108545-BORNARE HARSHAD

Authorized By : U321-HEAD_202700-DEO MUKESH

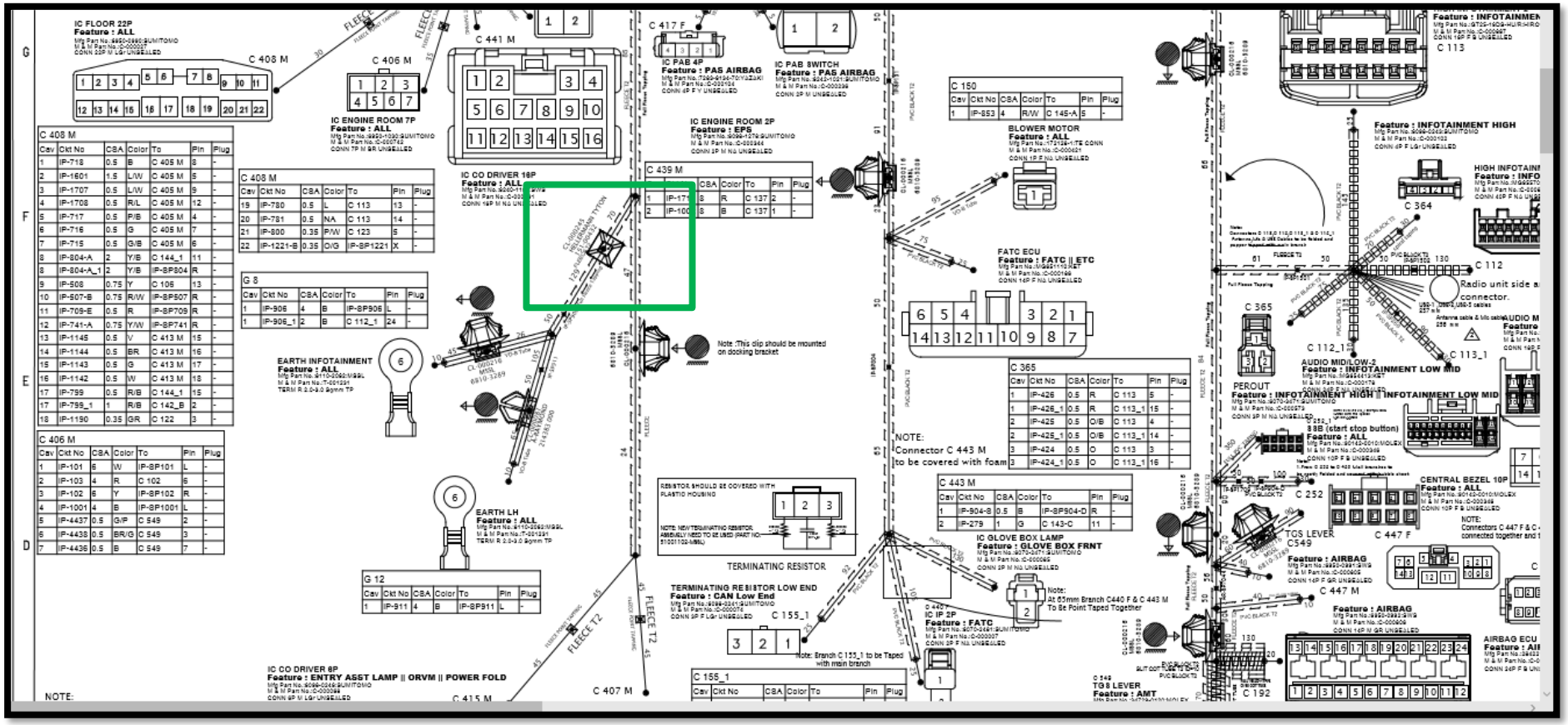
Legends

Add New : A Part/Assy that is getting released through ECN for the very first time

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

UPDATED DRAWING



8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

PFMEA

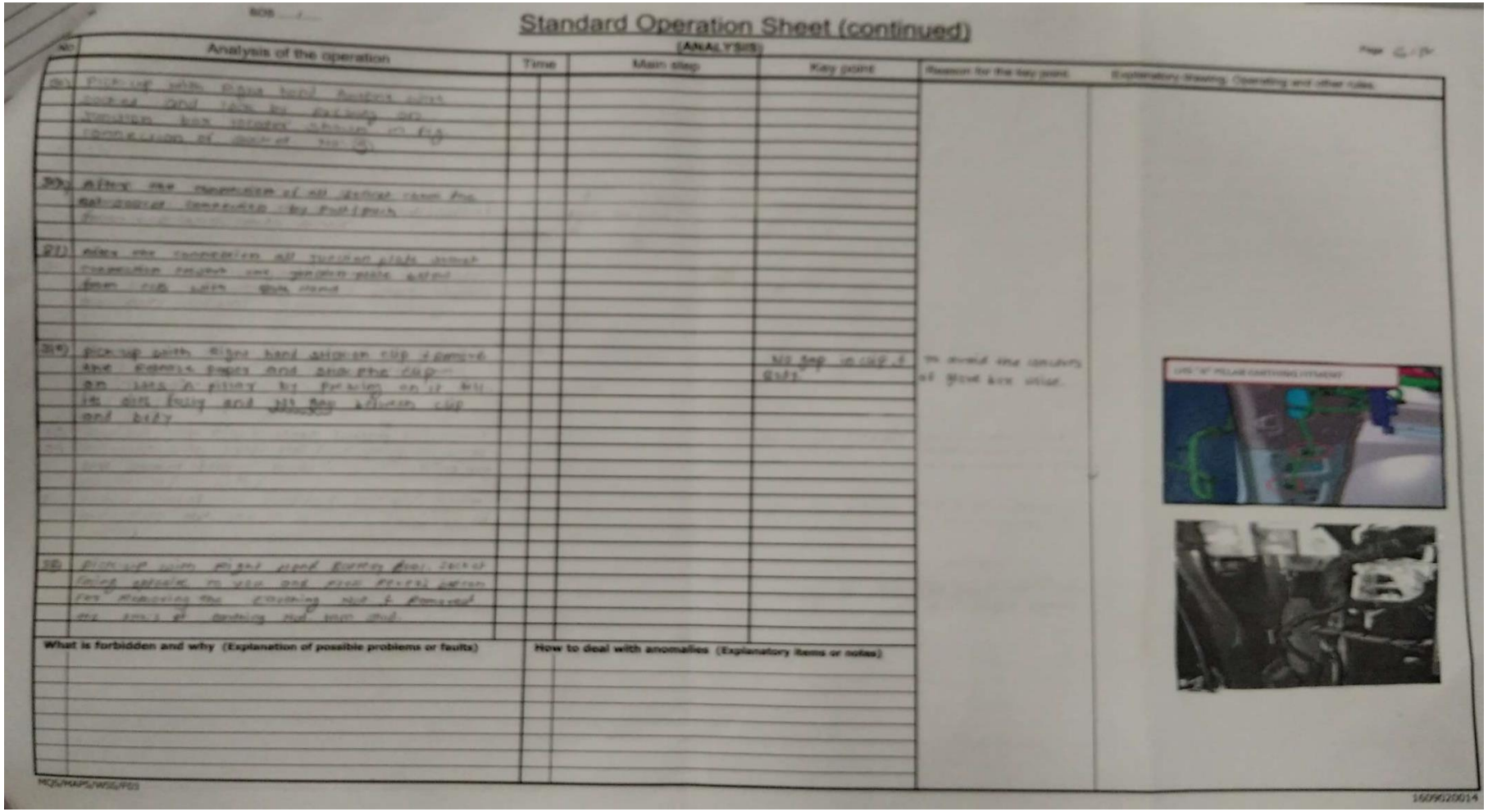
Plant : A003		Model :Marazzo														Cross Functional Team					
Part No/s.:		Part Name/ Process:	COCKPIT WIREHARNESS CONNECTION LH					Mahindra & Mahindra Ltd.								Vikrant Morade					
Latest Change Letter:		Optn No:	1801010					Process Potenial Failure Mode & Effect Analysis.								Yogesh Dhikale					
PU/ Shop :Scorpio, Marazzo TCF		Optn Desptn:	COCKPIT WIREHARNESS CONNECTION LH					Reviewed by :Amar Pande			Rev. No	6		Doc No.::MQS/PFMEA/NASHIK/U321/TCF/1801010		Vikrant Ghule					
Module/ Cell :TRIM LINE		M/C No.:						Approved by : Mandar Joshi			Rev. Dat	6.8.2019		FMEA Date (Orig) :08.04.2016		Sandip Joshi					
Process Step Function	Requirement	Potential Failure Mode	Potential Effect(s) of Failures	sev	las	Potential Cause(s)/Mechanism(s) of Failures	Current Process Controls Prevention	ccu	Current Process Controls Detection	etec	RPN	Recommended Action(s)	Responsibility & Target Completion Date	Action Results: Action Taken	Sev	Occ	Det	RPN	
ALIGN & PARK LH SIDE CONNECTOR FRAME ON BIW STUDS	SELECTION OF CONNECTOR	WRONG PART SELECTION	FURTHER OPERATION NOT POSSIBLE	4		OPERATOR MISTAKE	DESIGN POKA YOKE , FURTHER FITMENT NOT POSSIBLE	2	VISUAL INSPECTION	3	24										
	ALIGNMENT OF FRAME WITH STUDS	WRONG POSITIONING	FURTHER OPERATION NOT POSSIBLE	4		OPERATOR MISTAKE	OPERATOR TRAINING	3	SELF CHECK	7	84										
CONNECT ANTENNA, ENGINE ROOM, CO-DRIVER, FLOOR, ROOF WIRING ON PARKING FRAME AS SHOWN	CONNECTION OF CONNECTORS ON PARKING FRAME	IMPROPER LOCKING	End customer:-customer dissatisfaction (Ele. Function not work/malfunction) Internal customer:- rework	4		OPERATOR MISTAKE	DESIGN POKA YOKE , FURTHER FITMENT NOT POSSIBLE	2	VISUAL INSPECTION	7	70										
ENSURE CLICK NOISE OF CONNECTORS FOR LOCKING CONFORMATION	LOCKING CONFORMATION	IMPROPER LOCKING	End customer:-customer dissatisfaction (Ele. Function not work/malfunction) Internal customer:- rework	4		OPERATOR MISTAKE	OPERATOR TRAINING	2	VISUAL INSPECTION	7	112										
ENSURE FENCING CLIP PASTING	PASTING CONFIRMATION	IMPROPER PASTING	End customer : customer dissatisfaction(wire foul to glovebox lead to glove box noise)	4		OPERATOR MISTAKE	OPERATOR TRAINING	2	VISUAL INSPECTION	7	112										
TIGHTENING OF FRAME ON STUDS TO SPECIFIED TORQUE 6+/- 1 NM	PROPER TIGHTENING OF THE JOINT	TORQUE NOT ACHIEVED- UNDER TORQUE	End customer:-customer dissatisfaction Rattle noise from IP Internal customer:- rework	8		CALIBRATION MAY NOT HAVE DONE FOR THE REGULAR INTERVAL	(1)TORQUE CALIBRATION FREQUENCY TO BE DECIDE (2)HISTORY CARD OF THE TOOL TO BE MAINTAINED	2	AS PER AMC TOOL WILL BE CALIBRATED IN THE REGULAR INTERVALS	5	80										
						CALIBRATION SET VALUE IS WRONG	IF THE IDENTIFICATION LABEL IS MISSING NUTRUNNER SHOULD NOT BE USED	3	CALIBRATION RECORD	7	168										
						WRONG TOOL USED	DEDICATED TOOLS FOR SPECIFIED TORQUE AND LABELS ON THE TOOL	2	LABELS ON THE TOOL	7	112										
		TORQUE NOT ACHIEVED - OVER TORQUE	End customer:-customer dissatisfaction Rattle noise/thread dameged Internal customer:- rework	8		CLIBRATION SET VALUE IS WRONG	IF THE IDENTIFICATION LABEL IS MISSING NUTRUNNER SHOULD NOT BE USED	3	CALIBRATION RECORD	7	168										
						WRONG TOOL USED	DEDICATED TOOLS FOR SPECIFIED TORQUE AND LABELS ON THE TOOL	2	LABELS ON THE TOOL	7	112										

Rev No.:	Rev Date	Revision Summary	Reason for Revision
1	8/4/2016	Revision for VP-0	VP-0 Phase
2	7/18/2017	Revision for VP extension	VP Extension Phase
3	20.2.2018	VP Phase review	VP Phase
4	07.08.2018	Review no chenge	Six monthly revew
5	12.02.2019	Review no chenge	Six monthly revew
6	06.08.2019	Chenge in Occ rating	Six monthly revew

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SOS



Standard Operation Sheet (continued)

(ANALYSIS)

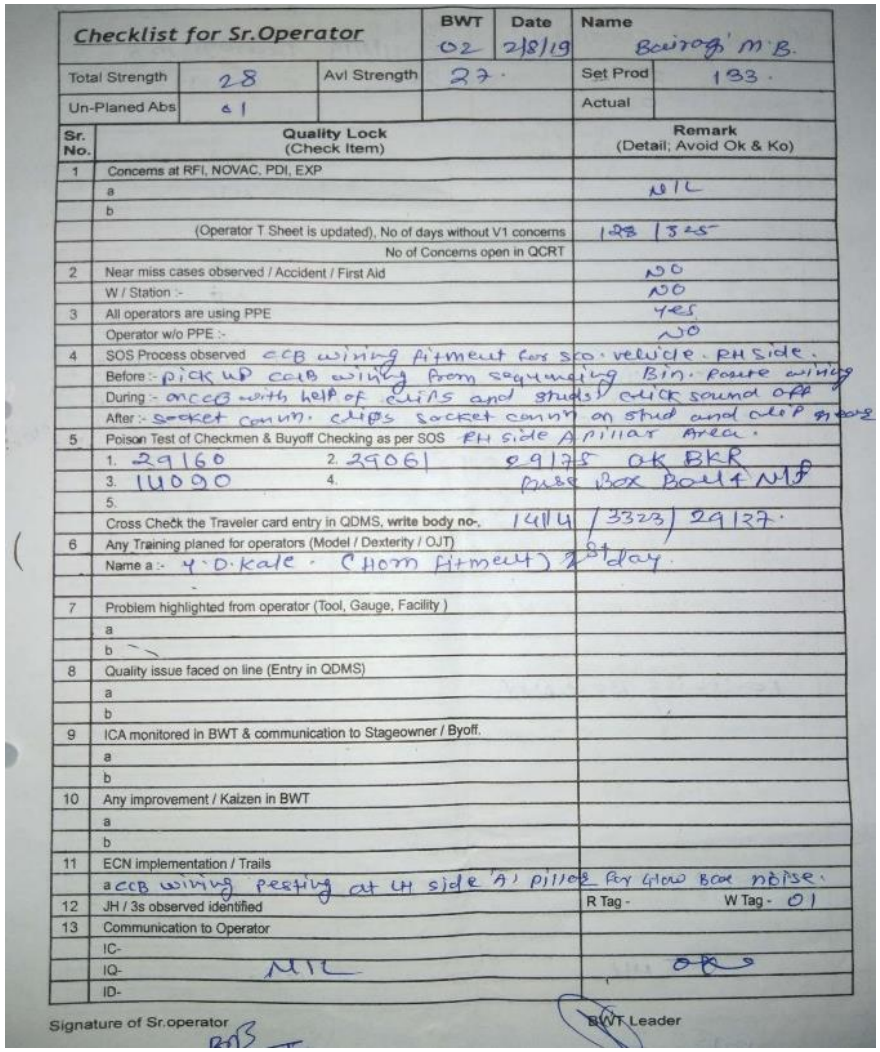
No	Analysis of the operation	Time	Main step	Key point	Reason for the key point	Explanatory drawing, Operating and other rules
30)	[illegible]					
31)	[illegible]					
32)	pick up with right hand [illegible] cup [illegible] and place the cup on [illegible]			No gap in cup & [illegible]	To avoid the [illegible] of glue [illegible]	
33)	pick up with right hand [illegible]					

What is forbidden and why (Explanation of possible problems or faults)

How to deal with anomalies (Explanatory items or notes)

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.



Checklist for Sr.Operator

BWT	Date	Name
02	2/8/19	Bairagi M.B.

Total Strength	28	Avl Strength	27	Set Prod	133
Un-Planed Abs	01			Actual	

Sr. No.	Quality Lock (Check Item)	Remark (Detail; Avoid Ok & Ko)
1	Concerns at RFI, NOVAC, PDI, EXP	
	a	NIL
	b	
	(Operator T Sheet is updated), No of days without V1 concerns	128 / 345
	No of Concerns open in QCRT	
2	Near miss cases observed / Accident / First Aid	NO
	W / Station :-	NO
3	All operators are using PPE	yes
	Operator w/o PPE :-	NO
4	SOS Process observed CCB wiring fitment for sro. vehicle. RH side.	
	Before :- pick up CCB wiring from segregating Bin. Route wiring	
	During :- on CCB with help of clips and studs click sound off	
	After :- socket conn. clips socket conn on stud and check [illegible]	
5	Poison Test of Checkmen & Buyoff Checking as per SOS RH side A pillar Area.	
	1. 29160 2. 29061 3. 29175 OK BKR	
	3. 14090 4. Fuse Box Bolt NIL	
	5.	
	Cross Check the Traveler card entry in QDMS, write body no-. 1414 / 3323 / 29127.	
6	Any Training planed for operators (Model / Dexterity / OJT)	
	Name a :- Y.D. Kale. (Horn fitment) 2nd day.	
7	Problem highlighted from operator (Tool, Gauge, Facility)	
	a	
	b	
8	Quality issue faced on line (Entry in QDMS)	
	a	
	b	
9	ICA monitored in BWT & communication to Stageowner / Byoff.	
	a	
	b	
10	Any improvement / Kaizen in BWT	
	a	
	b	
11	ECN implementation / Trails	
	a CCB wiring pasting at LH side 'A' pillar for Glow Box noise.	
12	JH / 3s observed identified	R Tag - W Tag - 01
13	Communication to Operator	
	IC-	
	IQ- NIL	OK
	ID-	

Signature of Sr.operator

BWT Leader

S.O.Check sheet

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

QA MATRIX

QA MATRIX

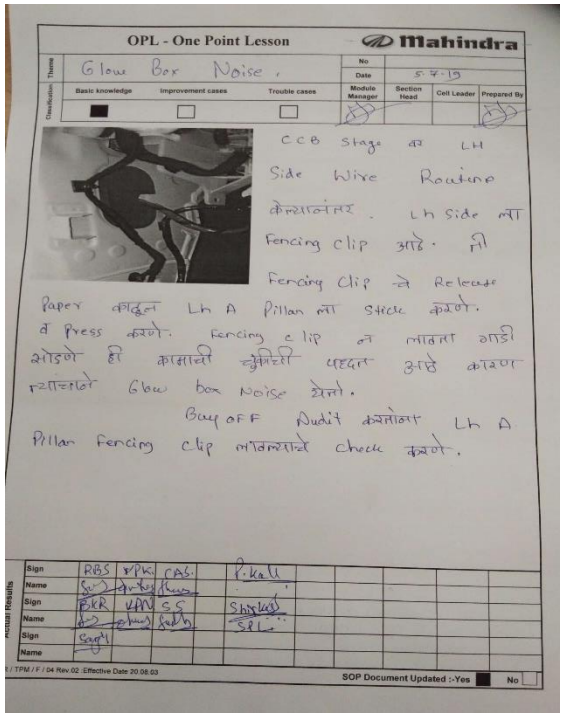
PU NAME	SCORPIO	INITIALIZATION DATE		Global Quality Guarantee level.	
DEPT NAME	SXTCF	SIGNATURES			
SECTION NAME	TRIM1	Dept Head	Rahul narkar	Section	100
		Module Manager	SANDEEP CHIKHLE	Plant	100
		Section Manager	2 SP	Revision date	

Workstation No.	Process description (POS)	What To Check	Date	Failure modes	Severity	Prev. occurence	Potential decfect	Inspection process: Section						Inspection process: Outside section				Guarantee value				Faults recorded			Quality guarantee level				Comments
								plugging	latch	wiring	roof	ccb	abs	by off	FAI	CAI	Trim			Workstation	Section	Outside section	PLANT	Section	Outside section	CUSTOMER	WORKSTATION	Section	Outside section	PLANT	
																								1M	1M	3M					
	Glove box Noise	CCB wire routine	5.7.19	Fouling	3							⊙ 3		▯ 1		▯ 1				3	4	1	5				OK	OK	NOK	Ok	

8. Standardization

8.2 Training:

SN	Training Identified	Actual date	Responsibility	Status
1.	Awareness given through OPL.	5.7.2019	Prasad/Mahesh	Closed



OPL - One Point Lesson

Mahindra

Theme: Glow Box Noise.

No

Date: 5.7.19

Classification: Basic knowledge | Improvement cases | Trouble cases

Module Manager | Section Head | Cell Leader | Prepared By

CCB Stage वर LH Side Wire Routing केल्यानंतर. LH Side ला Fencing clip आहे. ही Fencing Clip ने Release Paper काढून LH A Pillar ला Stick करणे. व Press करणे. Fencing clip ने लावता गाडी शेडणे हे कामाची चुकीची पद्धत आहे कारण त्यामुळे Glow box Noise येतो.

Buy off Audit करताना LH A Pillar Fencing clip लावलयाचे check करणे.

Sign	RBS	VPK	CAG	P.Kall					
Name									
Sign	PKR	VAM	CS						
Name									
Sign									
Name									

Actual Results

/ TPM / F / 04 Rev.02 :Effective Date 20.08.03

SOP Document Updated :-Yes No

8. Standardization

8.3 Additional Audit & check

1	Check point added in buy off check sheet	5.7.19	Prasad Deore	Closed

TRIM-1 BUY-OFF - CHECK POINTS

				Rev no:-167	*Revision on - 16.09.19*
SC CC	**Description**	**What to Check**	**V1/V2**	**How to Check**	**Source of complaint**
CC/Z	Door striker screw	No loose fitment of door striker. No fouling of striker	V1	Marking	
CC/SC	CCB mounting & wiring	Full tightening of bolt & socket connection. Stripping of harness with CCB near brake pedal. Airbag wiring in hook wth strap Check for protective sleeve on CCB-U321 only Check 2 extra socket at LH fendor for domestic BMT Check elay mtg. on fuse box . Check extra socket at LH side for all domestic models . Ensure fencing clip pasted on LH A pillar	V1	Visual marking on relay marking for socket rpesent Marking on body near clip for Marazzo pull/push marking on ignition socket.	RFI CAI

Page 3

9. Summarize & Plan future actions

9.1 Conclusion:

Glove box noise concern eliminated by correction in process & part improvements.

9.2 Horizontal Deployment:

Analysis is useful in upcoming projects, inputs to DFMEA.

9.3 Lessons Learnt:

Things Gone Wrong :

Initially only mfg process was suspected as cause of concern. Wire fouling concern not considered while product design.

Things Gone Right :

Brainstorming Happened between CFT team & problem solved by QC story approach.

.

Thank you

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